

LiFT Installation Manual for Windows

1. Overview

This manual describes how to install **LiFT (Live Foci Tracking)** on a Windows computer.

LiFT is a software package for live-cell microscopy analysis, including foci detection and tracking. The official project resources are:

- Python package <https://pypi.org/project/live-foci/>
 - LiFT GitHub repository <https://github.com/UU-cellbiology/LiveFoci/>
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2. System requirements

The following instructions are intended for **Windows 10/11**.

The installation uses:

- Python 3.12
- Microsoft Visual Studio Build Tools
- A Python virtual environment
- LiFT and its optional dependencies

Important: *Python 3.12 is recommended for this installation. Do not use Python 3.14 for this setup, as some of the scientific-computing dependencies used by LiFT contain compiled components that may not yet be compatible with Python 3.14.*

3. Install Microsoft Visual Studio Build Tools

Some of LiFT's dependencies contain compiled C/C++ components. Windows therefore needs the Microsoft C++ build tools.

- Download the installer: https://aka.ms/vs/stable/vs_BuildTools.exe
- Run the installer, select the required components:
 - Desktop development with C++
 - MSVC v14.x C++ build tools
 - Windows 10/11 SDK

You do not need to install the full Visual Studio IDE. The **Build Tools** installation is sufficient. Complete the installation before continuing.

4. Install Python 3.12

To get python installed you have two options:

1. Open the Microsoft Store, search for "Python Install Manager" and install it. After installation, restart PowerShell. In the opened Command Prompt run:

```
py install 3.12
```

or

2. Go to python.org and download the windows installer for version for the latest 3.12.x and run it. During setup, tick "Add python.exe to PATH" and leave "py launcher" checked.

after completing one of the previous steps, verify that Python 3.12 is available:

```
py -3.12 --version
```

You should see something similar to:

```
Python 3.12.x
```

5. Create a Python virtual environment

Create a dedicated virtual environment for LiFT:

```
py -3.12 -m venv C:\Users\<USERNAME>\lift_cp312
```

Replace <USERNAME> with the Windows username of the user installing LiFT.

6. Activate the virtual environment

From Command Prompt, activate the environment:

```
.\lift_cp312\Scripts\activate
```

After activation, the command prompt should show the environment name, for example:

```
(lift_cp312) C:\Users\...
```

You can verify that the correct Python is being used:

```
python --version
```

7. Update Python packaging tools

With the virtual environment activated, update pip, setuptools, and wheel:

```
python -m pip install --upgrade pip setuptools wheel
```

This ensures that the latest available packaging tools are used when installing LiFT and its dependencies.

8. Install LiFT

Install LiFT together with the required optional components:

```
python -m pip install "live-foci[notebook,app,cp-v3,spotiflow,trackastra]"
```

The installation may take several minutes because a number of scientific Python packages and machine-learning dependencies are installed.

9. Verify the LiFT installation

After installation, make sure the virtual environment is still active, then run:

```
lift info
```

A successful installation should produce an environment summary similar to:

— LiFT environment —		
✓ cellpose	unknown version	[cp-sam / cp-v3]
✓ trackastra	0.5.5	[trackastra]
✓ spotiflow	0.6.5	[spotiflow]
✓ torch	2.13.0+cpu	[cp-sam/cp-v3]

The exact versions may differ from those shown above as dependencies are updated.

Use the GUI

1. Download the required files from github

Download the templates, and the LiFT_app.py file from github and place them at the location of your preference on your computer.

2. Change the directory to move to the folder where you saved those files.

Now change the directory in PowerShell to the folder you downloaded (and unzipped) in the previous step. We placed it inside the document folder in a folder called "lift_package". Change the path in Powershell by using the following command:

```
cd Documents\lift_package
```

The new path in PowerShell should now be visible and look as:

```
(lift_cp312) C:\Users\<username>\Documents\lift_package
```

3. Launch the user interface

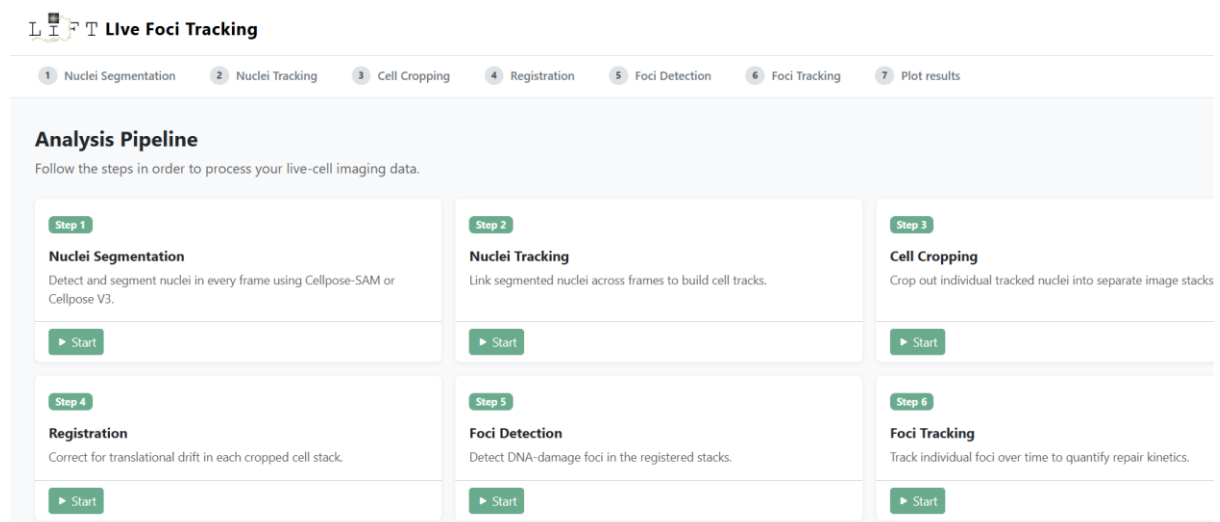
In the open command prompt we can now run the Python that allows you to use LiFT through the graphical user interface (GUI):

```
python LiFT_app.py
```

This should return some lines including a link: <http://127.0.0.1:5000>

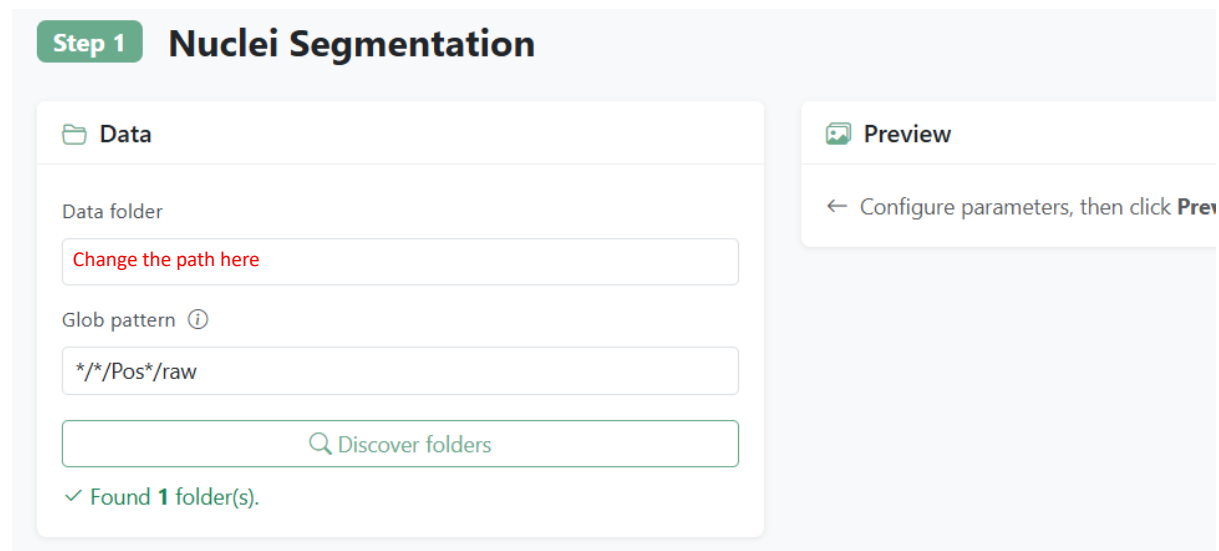
Open this link in a browser which should give you the (GUI).

The GUI should now be running and show you all the different steps:



Step 1: Nucleus segmentation using Cellpose

Your data should now be placed inside a data folder. Inside the GUI, there is a box where you can type where the data is stored. Change it to specify the folder with your data and press “Discover folders”. It should detect your data now:



Step 1 **Nuclei Segmentation**

Data

Data folder

Change the path here

Glob pattern ⓘ

//Pos*/raw

Discover folders

✓ Found 1 folder(s).

Preview

← Configure parameters, then click **Preview**

First run the segmentation using the “Preview” button at the bottom. It will get some random samples and display the segmentation results. Tune the parameters and press the button again to see the result with these new parameters. If you are satisfied with the results use the green button “Run segmentation on all data” at the bottom to process all data. Now continue to the other steps which work in a very similar fashion.